

Chapter 3: Observational Studies

Applied Use Case: Did the Retention Offer Actually Reduce Churn?

1 Introduction: Observational CRM Data vs. A/B Tests

In business applications, conducting randomized experiments (A/B tests) across the entire customer base is frequently infeasible due to budget constraints, ethical concerns, or risk of customer churn. Instead, companies must evaluate retention strategies using historical observational CRM data.

Chapter 3 of *Causal Inference: What If* addresses a central question: **Under what conditions can observational data be analyzed as if it arose from a conditionally randomized experiment?**

To answer this, we analyze:

- **Treatment (A):** $A = 1$ if customer received a retention offer; $A = 0$ if no offer.
- **Outcome (Y):** $Y = 1$ if customer churned within 60 days; $Y = 0$ if retained.
- **Measured Covariates (L):** Baseline churn risk factors (e.g., tenure, usage drop, complaint history).
- **Unmeasured Confounders (U):** Unrecorded customer attributes (e.g., verbal complaints to support agents).

2 The Three Nonparametric Identifiability Conditions

For observational data to yield unbiased causal effect estimates (e.g., via Standardization or IP Weighting), three core **identifiability conditions** must hold simultaneously.

2.1 1. Conditional Exchangeability ($Y^a \perp\!\!\!\perp A \mid L$)

In historical CRM data, treatment A was assigned non-randomly: dissatisfied or high-risk customers ($L = 1$) received offers far more frequently than low-risk customers ($L = 0$). Thus, treated and untreated groups are not marginally exchangeable:

$$\Pr[Y^{a=0} = 1 \mid A = 1] \neq \Pr[Y^{a=0} = 1 \mid A = 0]$$

Conditional exchangeability assumes that **within every stratum of measured covariates L** , treatment assignment is independent of potential outcomes:

$$Y^a \perp\!\!\!\perp A \mid L \quad \text{for } a \in \{0, 1\}$$

Unmeasured Confounding (U)

If there exists an unmeasured variable U (e.g., an offline complaint) that influences both treatment assignment A and churn outcome Y , conditional exchangeability fails ($Y^a \not\perp\!\!\!\perp A \mid L$), resulting in residual confounding bias.

2.2 2. Positivity ($0 < \Pr[A = a \mid L = l] < 1$)

The positivity condition (also known as the *experimental treatment assumption*) requires that every customer segment defined by L has a non-zero probability of receiving every treatment level:

$$0 < \Pr[A = a \mid L = l] < 1 \quad \text{for all } a \text{ and } l \text{ where } \Pr[L = l] > 0$$

- **Deterministic Policy Violation:** If company policy dictates that high-complaint customers *always* receive discounts ($\Pr[A = 1 \mid L = \text{high}] = 1$) or low-risk customers *never* receive offers ($\Pr[A = 1 \mid L = \text{low}] = 0$), positivity is structurally violated.
- **Consequence:** Observational data contains zero counterfactual information for unobserved cells, making standardization and IP weighting ill-defined.

2.3 3. Consistency ($Y = Y^A$ & Well-Defined Interventions)

The consistency assumption links potential outcomes to observed data through two requirements:

1. **Observed Link:** An individual's observed outcome equals their potential outcome under the treatment actually received:

$$Y = Y^A = AY^{a=1} + (1 - A)Y^{a=0}$$

2. **Sufficiently Well-Defined Intervention:** Treatment A must correspond to a specific, unambiguous operational action.

Treatment Variation Irrelevance in Churn Analysis

If “retention offer” ($A = 1$) includes multiple versions in historical CRM data—such as a 10% discount, a 50% discount, or an upgraded subscription—the counterfactual outcome $Y^{a=1}$ is ill-defined. To ensure consistency, the intervention must be defined precisely (e.g., “A 20% discount on monthly subscription for 3 months”).

3 Why Identifiability Assumptions Are Untestable

A foundational principle emphasized in Chapter 3 is that **identifiability assumptions cannot be verified or proven from data alone**:

- **Exchangeability:** Requires knowing $\Pr[Y^{a=1} = 1 \mid A = 0]$, which involves missing counterfactual outcomes for untreated customers. Data cannot verify whether unmeasured confounders U exist.
- **Consistency:** Cannot be verified because we never observe both $Y^{a=1}$ and $Y^{a=0}$ for the same customer to verify treatment variation irrelevance.
- **Positivity:** Can be partially checked for measured covariates L by examining cell counts, but cannot be verified for unmeasured confounders U .

Causal inference from observational studies always requires combining empirical data with **substantive domain expertise**.

4 Target Trial Emulation Framework

To prevent common observational study biases (such as immortal time bias or ill-defined interventions), Hernán and Robins introduce **Target Trial Emulation**:

1. **Design the Target Trial**: Specify the protocol of a hypothetical ideal A/B test (eligibility, treatment arms, assignment mechanism, time zero, follow-up, outcome).
2. **Emulate the Protocol**: Map each component of the target trial onto the observational CRM database using g-methods (Standardization / IPW).

Table 1: Target Trial Emulation Protocol for Retention Offer Use Case

Protocol Component	Hypothetical Target Trial	Observational CRM Emulation
Eligibility Criteria	Active subscribers with tenure ≥ 3 mos.	Same CRM filter at index date T_0
Interventions	$A = 1$: 20% discount for 3 months $A = 0$: Standard pricing	Identify exact offer $A = 1$ vs $A = 0$ recorded in CRM logs
Assignment	Randomized at T_0	Emulated via IPW / Standardization (L)
Time Zero (T_0)	Day of randomization	Day eligibility and treatment recorded
Outcome	Subscription cancellation within 60 days	60-day churn flag in CRM billing

5 Summary of Key Formulae

When the three identifiability conditions hold, observational data can be used to estimate population causal effects via:

Standardization (Parametric g-Formula) : $\Pr[Y^a = 1] = \sum_l \Pr[Y = 1 \mid A = a, L = l] \cdot \Pr[L = l]$

Inverse Probability Weighting (IPW) : $W^A = \frac{1}{f(A \mid L)}, \quad \Pr[Y^a = 1] = E \left[\frac{I(A = a)Y}{f(A \mid L)} \right]$